ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2020 P2HR Worked Solutions

Jan 2020 | P2HR

33 questions ▪ question image followed by worked solution

PREPARED BY

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION Q#: 1 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

Jan 2020 P2HR - Jan 2020 | P2HR

1 (a) Write $5^{17} \times 5^{2}$ as a single power of 5

....

(1)

(b) Write 800 as a product of its prime factors.
Show your working clearly.

....

(2)

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

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Topic check. Prime factors, HCF and LCM. The tag is correct.

Method

Use the index law $a^m \times a^n = a^{m+n}$:

$$5^{17} \times 5^2 = 5^{19}$$

Now write 800 as a product of prime factors:

$$800 = 8 \times 100$$

$$800 = 2^3 \times 2^2 \times 5^2$$

$$800 = 2^5 \times 5^2$$

FINAL ANSWER

(a) 5^{19}, (b) $2^5 \times 5^2$

PAST PAPER QUESTION

Q#: 2 Marks: 3

TOPIC: **Statistics Toolkit**

Jan 2020 P2HR - Jan 2020 | P2HR

2 The table gives information about the amount of money, in £, that Fiona spent in a grocery store each week during 2019

Amount spent (£x)	**Frequency**
$0 \leqslant x < 20$	5
$20 \leqslant x < 40$	11
$40 \leqslant x < 60$	8
$60 \leqslant x < 80$	19
$80 \leqslant x < 100$	9

Work out an estimate for the total amount of money that Fiona spent in the grocery store during 2019

£..

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 3

TOPIC: **Statistics Toolkit**

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Topic check. Statistics Toolkit. The tag is correct.

Solution

Use midpoints $10, 30, 50, 70, 90$.

$$10(5) + 30(11) + 50(8) + 70(19) + 90(9)$$

$$= 50 + 330 + 400 + 1330 + 810 = 2920$$

FINAL ANSWER

£2920.

PAST PAPER QUESTION

Q#: 3 Marks: 4

TOPIC: **Forming & Solving Equations**

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3 Three tins, A, B and C, each contain buttons.

Tin A contains x buttons.
Tin B contains 4 times the number of buttons that tin A contains.
Tin C contains 7 fewer buttons than tin A.

The total number of buttons in the three tins is 137

Work out the number of buttons in tin C.

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 4

TOPIC: **Forming & Solving Equations**

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Topic check. Corrected to Forming & Solving Equations.

Why it moved: The question asks us to form and solve an equation for the numbers of buttons in three tins. There is no probability event.

Solution

Let the number of buttons in tin A be x.
Then tin B has $4x$ buttons and tin C has $x - 7$ buttons.

$$x + 4x + (x - 7) = 137$$

$$6x - 7 = 137$$

$$6x = 144$$

$$x = 24$$

Tin C has

$$24 - 7 = 17$$

FINAL ANSWER
17 buttons.

PAST PAPER QUESTION

Q#: 4 Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2020 P2HR - Jan 2020 | P2HR

4 The diagram shows a rectangle and a diagonal of the rectangle.

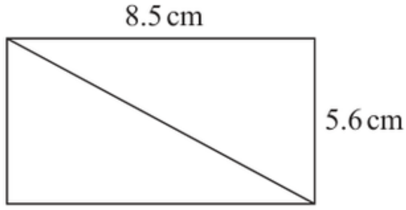

Diagram **NOT** accurately drawn

Work out the length of the diagonal of the rectangle.
Give your answer correct to 1 decimal place.

.... cm

(Total for Question 4 is 3 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

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Topic check. Corrected to Right-Angled Triangles - Pythagoras & Trigonometry.

Solution

$$d^2 = 8.5^2 + 5.6^2$$

$$d = \sqrt{103.61} = 10.178\ldots$$

FINAL ANSWER

10.2 cm.

PAST PAPER QUESTION

Q#: 5 Marks: 3

TOPIC: **Standard & Compound Units**

Jan 2020 P2HR - Jan 2020 | P2HR

5 A plane takes 3 hours 36 minutes to fly from the Cayman Islands to New York.
The plane flies a distance of 2470 km.

Work out the average speed of the plane in km/h.
Give your answer correct to the nearest whole number.

.... km/h

(Total for Question 5 is 3 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 3

TOPIC: **Standard & Compound Units**

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Topic check. Corrected to Standard & Compound Units.

Solution

The time is

$$3 \text{ h } 36 \text{ min} = 3.6 \text{ h}$$

Average speed:

$$2470 \div 3.6 = 686.111\ldots$$

FINAL ANSWER

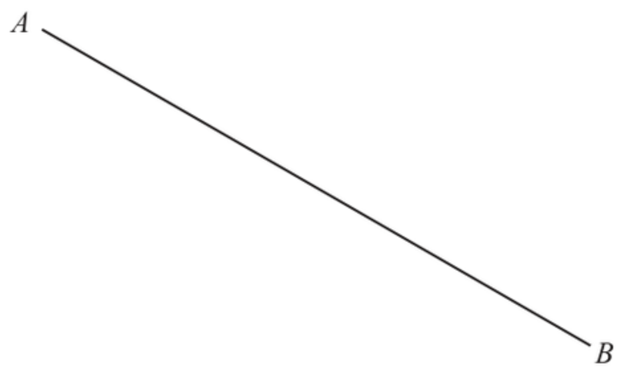
686 km/h.

PAST PAPER QUESTION Q#: 6 Marks: 2

TOPIC: **Bearings, Scale Drawing & Constructions**

Jan 2020 P2HR - Jan 2020 | P2HR

6 Use ruler and compasses only to construct the perpendicular bisector of the line AB.
You must show all your construction lines.



(Total for Question 6 is 2 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 2

TOPIC: **Bearings, Scale Drawing & Constructions**

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Topic check. Bearings, Scale Drawing & Constructions. The tag is correct.

Construction answer

To construct the perpendicular bisector of AB:

1. Set the compass radius to more than half of AB.
2. With centre A, draw arcs above and below the line.
3. With the same radius and centre B, draw arcs crossing the first pair.
4. Draw the straight line through the two arc intersections.

FINAL ANSWER

This line is the perpendicular bisector of AB.

PAST PAPER QUESTION

Q#: 7 **Marks: 3**

TOPIC: **Simultaneous Equations**

Jan 2020 P2HR - Jan 2020 | P2HR

7 Solve the simultaneous equations

$$\begin{aligned} 3x + 5y &= 6 \\ 7x - 5y &= -11 \end{aligned}$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 3

TOPIC: **Simultaneous Equations**

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Topic check. Simultaneous equations. The tag is correct.

Solution

$$3x + 5y = 6$$

$$7x - 5y = -11$$

Add the equations:

$$10x = -5$$

$$x = -\frac{1}{2}$$

Substitute into $3x + 5y = 6$:

$$3\left(-\frac{1}{2}\right) + 5y = 6$$

$$-\frac{3}{2} + 5y = 6$$

$$5y = \frac{15}{2}$$

$$y = \frac{3}{2}$$

FINAL ANSWER

$x = -\frac{1}{2}$, $y = \frac{3}{2}$.

PAST PAPER QUESTION

Q#: 8 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jan 2020 P2HR - Jan 2020 | P2HR

8 Hamish buys a new car for \$20 000
The car depreciates in value by 19% each year.

Work out the value of the car at the end of 3 years.
Give your answer to the nearest \$.

\$..

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

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Topic check. Compound Interest and Depreciation. The tag is correct.

Method

A depreciation of 19% gives a multiplier of

$$1 - 0.19 = 0.81$$

After 3 years,

$$20000(0.81)^3 = 10628.82$$

Correct to the nearest dollar,

$$10628.82 \approx 10629$$

FINAL ANSWER

$10629

PAST PAPER QUESTION

Q#: 9 Marks: 3

TOPIC: **Standard & Compound Units**

Jan 2020 P2HR - Jan 2020 | P2HR

9 The diagram shows a box in the shape of a cuboid.

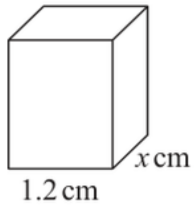

Diagram **NOT** accurately drawn

The box is put on a table.
The face of the box in contact with the table has length 1.2 metres and width x metres.

The force exerted by the box on the table is 27 newtons.
The pressure on the table due to the box is 30 newtons/m^2

$$\boxed{\text{pressure} = \frac{\text{force}}{\text{area}}}$$

Work out the value of x.

$x =$ ....

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Standard & Compound Units**

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WORKED SOLUTION

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Topic check. Corrected to Standard & Compound Units.

Solution

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$30 = \frac{27}{\text{area}}$$

$$\text{area} = \frac{27}{30} = 0.9 \text{ m}^2$$

The base area is

$$1.2x = 0.9$$

$$x = \frac{0.9}{1.2} = 0.75$$

FINAL ANSWER
$x = 0.75$ m.

PAST PAPER QUESTION Q#: 10 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

Jan 2020 P2HR - Jan 2020 | P2HR

10 The table shows information about the surface area of each of the world's oceans.

Ocean	**Surface area in square kilometres**
Pacific	1.56×10^8
Indian	6.86×10^7
Southern	2.03×10^7
Arctic	1.41×10^7
Atlantic	1.06×10^8

(a) Work out the difference, in square kilometres, between the surface area of the Atlantic Ocean and the surface area of the Indian Ocean.
Give your answer in standard form.

.... square kilometres
(2)

The surface area of the Pacific Ocean is k times the surface area of the Arctic Ocean.

(b) Work out the value of k.
Give your answer correct to the nearest whole number.

$k =$ ....
(1)

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

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Topic check. Powers, roots and standard form. The tag is correct.

Method

Atlantic surface area:

$$1.06 \times 10^8 = 106000000$$

Indian surface area:

$$6.86 \times 10^7 = 68600000$$

Difference:

$$106000000 - 68600000 = 37400000$$

$$37400000 = 3.74 \times 10^7$$

For part (b),

$$k = \frac{1.56 \times 10^8}{1.41 \times 10^7}$$

$$k = 11.063\ldots$$

To the nearest whole number,

$$k = 11$$

FINAL ANSWER

(a) 3.74×10^7, (b) 11

PAST PAPER QUESTION

Q#: 11 **Marks: 4**

TOPIC: **Graphing Inequalities**

Jan 2020 P2HR - Jan 2020 | P2HR

11 (a) Write down the integer values of x that satisfy the inequality $-2 < x \leqslant 4$

....

(2)

The region **R**, shown shaded in the diagram, is bounded by three straight lines.

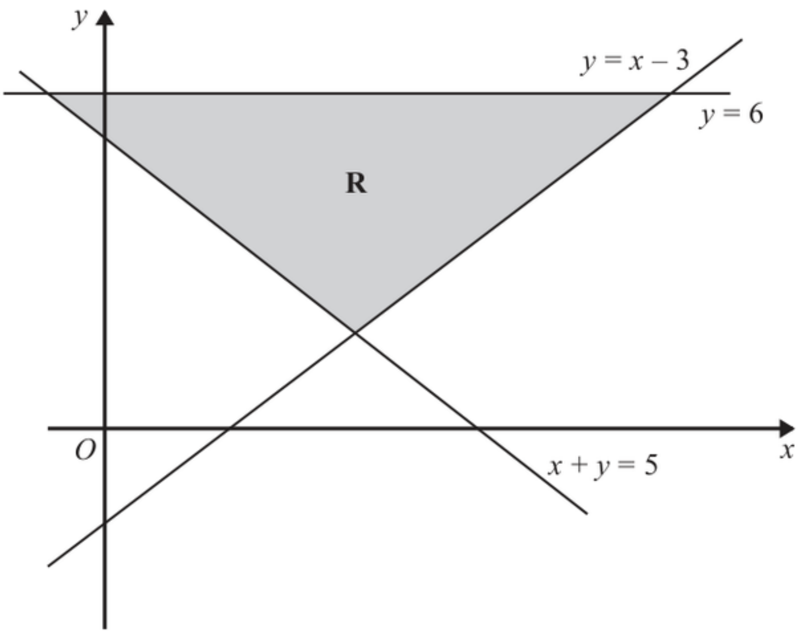

Diagram **NOT** accurately drawn

(b) Write down the three inequalities that define the region **R**.

....

(2)

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 4

TOPIC: **Graphing Inequalities**

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Topic check. Correct. This combines integer inequalities and a shaded region.

Solution

For part (a),

$$2 < x \leq 4$$

The integer values are

$$3, 4$$

For the region R, the boundaries are

$$y = x - 3, \qquad x + y = 5, \qquad y = 6$$

The shaded region is above the two sloping lines and below the horizontal line:

$$y \geq x - 3$$

$$x + y \geq 5$$

$$y \leq 6$$

FINAL ANSWER

(a) $3, 4$. (b) $y \geq x - 3$, $x + y \geq 5$, $y \leq 6$.

PAST PAPER QUESTION Q#: 12 Marks: 3

Topic: **Congruence, Similarity & Geometrical Proof**

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12 The diagram shows two congruent isosceles triangles and parts of two congruent regular polygons, **X** and **Y**.

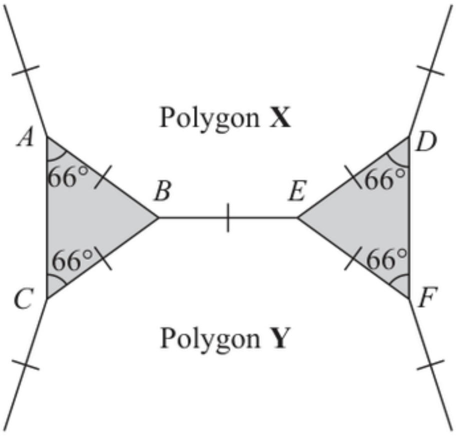

Diagram **NOT** accurately drawn

The two regular polygons each have n sides.

Work out the value of n.

$n =$ ....

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 3

TOPIC: **Congruence, Similarity & Geometrical Proof**

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Topic check. Congruence, Similarity & Geometrical Proof. The tag is correct.

Solution

Each congruent isosceles triangle has two angles of $66°$.
So the angle at the point between the two polygons is

$$180° - 66° - 66° = 48°$$

The two regular polygons have equal interior angles. Around the point:

$$48° + I + I = 360°$$

$$2I = 312°$$

$$I = 156°$$

The exterior angle is

$$180° - 156° = 24°$$

$$n = \frac{360}{24} = 15$$

FINAL ANSWER

$n = 15$.

PAST PAPER QUESTION Q#: 13 Marks: 3

TOPIC: **Volume & Surface Area**

Jan 2020 P2HR - Jan 2020 | P2HR

13

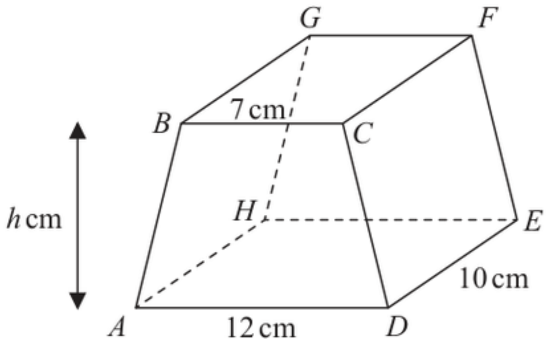

Diagram **NOT** accurately drawn

The diagram shows a prism *ABCDEFGH* in which *ABCD* is a trapezium with *BC* parallel to *AD* and *CDEF* is a rectangle.

$BC = 7\,\text{cm}$ $AD = 12\,\text{cm}$ $DE = 10\,\text{cm}$

The height of trapezium *ABCD* is h cm
The volume of the prism is $608\,\text{cm}^3$

Work out the value of h.

$h =$ ....

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 3

TOPIC: **Volume & Surface Area**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

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Topic check. Corrected to Volume & Surface Area.

Solution

Area of the trapezium cross-section:

$$\frac{1}{2}(7+12)h = 9.5h$$

Volume of the prism:

$$9.5h \times 10 = 608$$

$$95h = 608$$

$$h = 6.4$$

FINAL ANSWER

$h = 6.4$ cm.

PAST PAPER QUESTION

Topic: **Statistics Toolkit**

Jan 2020 P2HR - Jan 2020 | P2HR

Q#: 14 Marks: 3

14 Max kept a record of the marks he scored in each of the 11 spelling tests he took one term.

Here are his marks.

18 5 7 12 11 18 15 16 17 13 14

Find the interquartile range of the marks.

(Total for Question 14 is 3 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 3

TOPIC: **Statistics Toolkit**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Statistics Toolkit. The tag is correct.

Solution

Put the marks in order:

$$5,\ 7,\ 11,\ 12,\ 13,\ 14,\ 15,\ 16,\ 17,\ 18,\ 18$$

$$Q_1 = 11, \qquad Q_3 = 17$$

$$\text{IQR} = 17 - 11 = 6$$

FINAL ANSWER

6.

PAST PAPER QUESTION Q#: 15 Marks: 4

TOPIC: **Graphs of Functions**

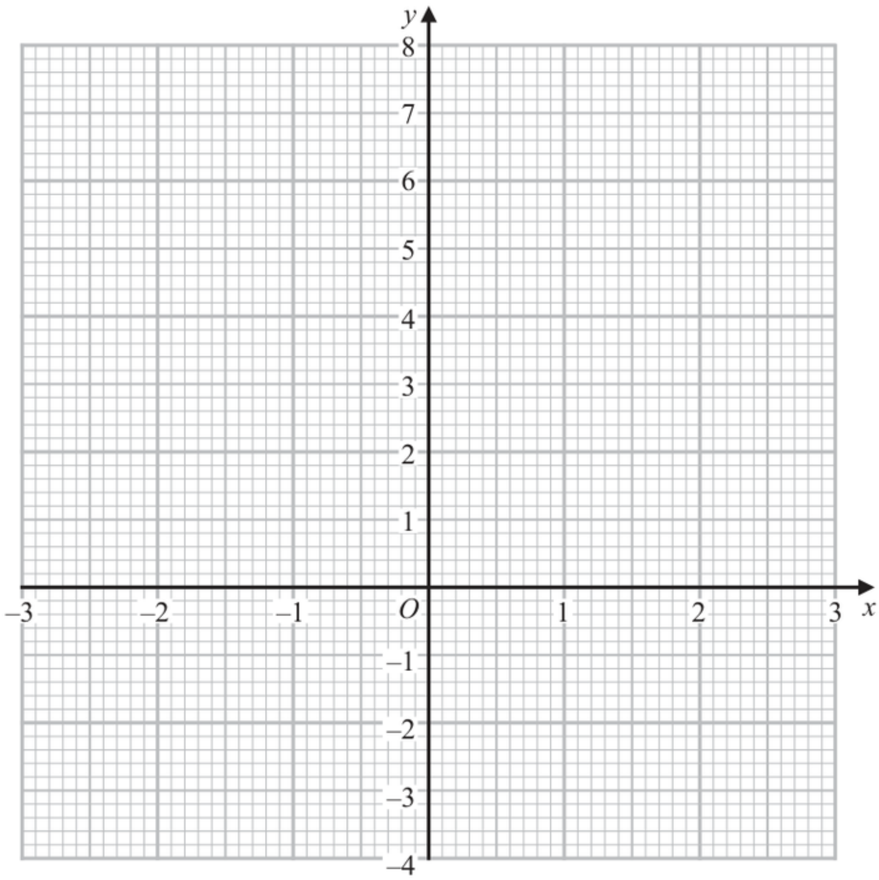
Jan 2020 P2HR - Jan 2020 | P2HR

15 (a) Complete the table of values for $y = x^2 - \frac{x}{2} - 3$

x	–3	–2	–1	0	1	2	3
y	7.5				–2.5		4.5

(2)

(b) On the grid, draw the graph of $y = x^2 - \frac{x}{2} - 3$ for values of x from –3 to 3



(2)

(Total for Question 15 is 4 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 4

TOPIC: **Graphs of Functions**

Jan 2020 P2HR - Jan 2020 | P2HR

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Topic check. Correct. This is a table-and-graph question for a quadratic function.

Solution

$$y = x^2 - \frac{x}{2} - 3$$

The completed table is:

x	-3	-2	-1	0	1	2	3
y	7.5	2	-1.5	-3	-2.5	0	4.5

Plot these points and join them with a smooth U-shaped curve.

FINAL ANSWER

Missing values are $2, -1.5, -3, 0$.

PAST PAPER QUESTION

Q#: 16 Marks: 8

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Jan 2020 P2HR - Jan 2020 | P2HR

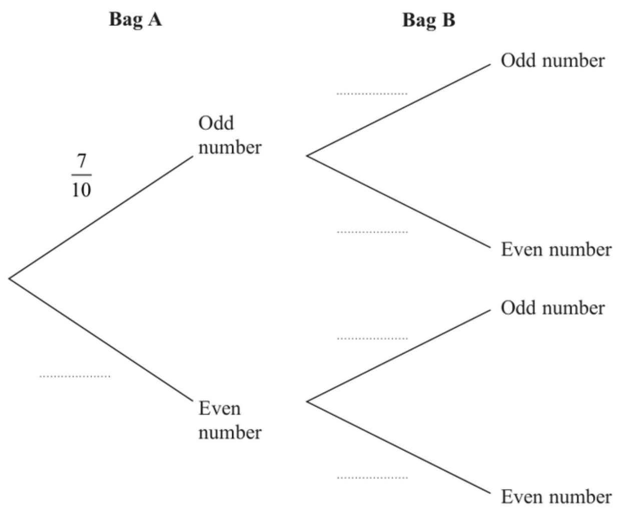
16 Cody has two bags of counters, bag **A** and bag **B**.

Each of the counters has either an odd number or an even number written on it.

There are 10 counters in bag **A** and 7 of these counters have an **odd** number written on them.
There are 12 counters in bag **B** and 7 of these counters have an **odd** number written on them.

Cody is going to take at random a counter from bag **A** and a counter from bag **B**.

(a) Complete the probability tree diagram.



(2)

(b) Calculate the probability that the total of the numbers on the two counters will be an odd number.

....

(3)

Harriet also has a bag of counters.
Each of her counters also has either an odd number or an even number written on it.

Harriet is going to take at random a counter from her bag of counters.

The probability that the number on each of Cody's two counters **and** the number on Harriet's counter will all be even is $\frac{3}{100}$

(c) Find the least number of counters that Harriet has in her bag.
Show your working clearly.

....

(3)

(Total for Question 16 is 8 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 8

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

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Topic check. Probability Diagrams - Venn & Tree Diagrams. The tag is correct.

Solution

For Cody:

$$P(A \text{ odd}) = \frac{7}{10}, \qquad P(A \text{ even}) = \frac{3}{10}$$

$$P(B \text{ odd}) = \frac{7}{12}, \qquad P(B \text{ even}) = \frac{5}{12}$$

An odd total comes from odd + even or even + odd:

$$\frac{7}{10} \times \frac{5}{12} + \frac{3}{10} \times \frac{7}{12} = \frac{7}{15}$$

Cody takes two even counters with probability

$$\frac{3}{10} \times \frac{5}{12} = \frac{1}{8}$$

Given Cody and Harriet both take two even counters:

$$\frac{1}{8} \times P(H \text{ even}) = \frac{3}{100}$$

$$P(H \text{ even}) = \frac{6}{25}$$

So Harriet's bag must have at least 25 counters.

FINAL ANSWER

$\frac{7}{15}$, and 25.

PAST PAPER QUESTION

Q#: 17 Marks: 6

TOPIC: **Set Notation & Venn Diagrams**

Jan 2020 P2HR - Jan 2020 | P2HR

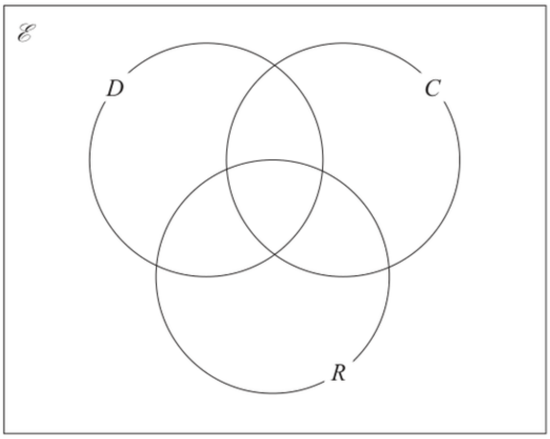
17 Some students in a school were asked the following question.

"Do you have a dog (D), a cat (C) or a rabbit (R)?"

Of these students

28 have a dog
18 have a cat
20 have a rabbit
8 have both a cat and a rabbit
9 have both a dog and a rabbit
x have both a dog and a cat
6 have a dog, a cat and a rabbit
5 have not got a dog or a cat or a rabbit

(a) Using this information, complete the Venn diagram to show the number of students in each appropriate subset.
Give the numbers in terms of x where necessary.



(3)

Given that a total of 50 students answered the question,

(b) work out the value of x.

x = ...

(2)

(c) Find $n(C' \cap D')$

....

(1)

(Total for Question 17 is 6 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 6

TOPIC: **Set Notation & Venn Diagrams**

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Topic check. Set notation and Venn diagrams. The tag is correct.

Method

Let the sets be D for dogs, C for cats, and R for rabbits.
The number in all three sets is 6.

$$D \cap R \text{ only} = 9 - 6 = 3$$

$$C \cap R \text{ only} = 8 - 6 = 2$$

$$D \cap C \text{ only} = x - 6$$

So the remaining one-set regions are

$$D \text{ only} = 25 - x, \quad C \text{ only} = 16 - x, \quad R \text{ only} = 9$$

There are 5 students outside all three sets. The total number of students is 50, so

$$(25 - x) + (16 - x) + 9 + (x - 6) + 3 + 2 + 6 + 5 = 50$$

$$60 - x = 50$$

$$x = 10$$

For $C' \cap D'$, we need students who are not in C and not in D. These are the rabbit-only students and the students outside all three sets.

$$9 + 5 = 14$$

FINAL ANSWER

$x = 10$ and $n(C' \cap D') = 14$

PAST PAPER QUESTION Q#: 18 Marks: 2

Topic: **Circle Theorems**

Jan 2020 P2HR - Jan 2020 | P2HR

18

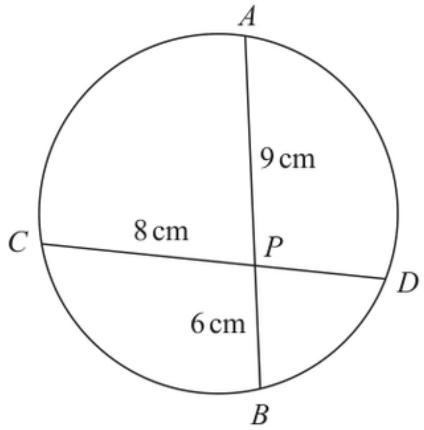

Diagram **NOT** accurately drawn

APB and *CPD* are chords of a circle.

$AP = 9$ cm $\qquad PB = 6$ cm $\qquad CP = 8$ cm

Calculate the length of *PD*.

.... cm

(Total for Question 18 is 2 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 2

TOPIC: **Circle Theorems**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

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Topic check. Circle Theorems. The tag is correct.

Solution

Use the intersecting chords theorem:

$$AP \times PB = CP \times PD$$

Substitute the values:

$$9 \times 6 = 8 \times PD$$

$$54 = 8PD$$

$$PD = 6.75$$

FINAL ANSWER

6.75 cm.

PAST PAPER QUESTION Q#: 19 Marks: 6

TOPIC: **Solving Inequalities**

Jan 2020 P2HR - Jan 2020 | P2HR

19 (a) Solve $\dfrac{4-3x}{5} - \dfrac{3x-5}{2} = -3$

Show clear algebraic working.

$x =$ ....

(3)

(b) Solve the inequality $5y^2 - 17y \leqslant 40$

....

(3)

(Total for Question 19 is 6 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 6

TOPIC: **Solving Inequalities**

Jan 2020 P2HR - Jan 2020 | P2HR

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Topic check. Solving inequalities. The tag is correct.

Solution

(a)

$$\frac{4-3x}{5} - \frac{3x-5}{2} = -3$$

Multiply by 10:

$$2(4-3x) - 5(3x-5) = -30$$

$$8 - 6x - 15x + 25 = -30$$

$$33 - 21x = -30$$

$$-21x = -63$$

$$x = 3$$

(b)

$$5y^2 - 17y \leq 40$$

$$5y^2 - 17y - 40 \leq 0$$

Factorise:

$$(5y+8)(y-5) \leq 0$$

The roots are $-\frac{8}{5}$ and 5. The quadratic is ≤ 0 between the roots.

FINAL ANSWER

$x = 3$, and $-\frac{8}{5} \leq y \leq 5$.

PAST PAPER QUESTION Q#: 20 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jan 2020 P2HR - Jan 2020 | P2HR

20 The diagram shows two similar vases, **A** and **B**.

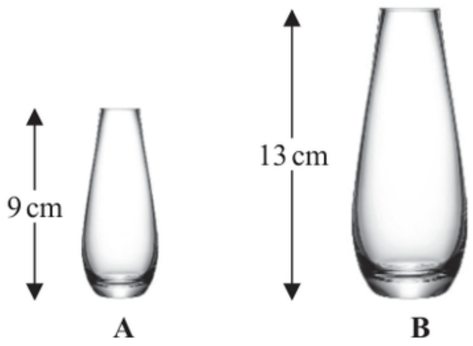

Diagram **NOT** accurately drawn

The height of vase **A** is 9 cm and the height of vase **B** is 13 cm.

Given that

surface area of vase **A** + surface area of vase **B** = 1800 cm^2

calculate the surface area of vase **A**.

.... cm^2

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The two vases are similar.
Their heights are in the ratio

$$9 : 13$$

So their surface areas are in the ratio

$$9^2 : 13^2 = 81 : 169$$

The total surface area is 1800 cm^2, so there are

$$81 + 169 = 250$$

parts altogether.
Surface area of vase A:

$$1800 \times \frac{81}{250} = 583.2$$

FINAL ANSWER

$\boxed{583.2 \text{ cm}^2}$

PAST PAPER QUESTION **Q#: 20** **Marks: 4**

TOPIC: **Area & Volume of Similar Shapes**

Jan 2020 P2HR - Jan 2020 | P2HR

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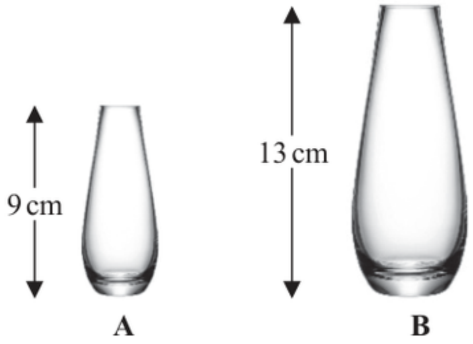

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(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

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FINAL ANSWER

$\boxed{583.2\ \text{cm}^2}$

PAST PAPER QUESTION Q#: 21 Marks: 7

TOPIC: **Algebraic Roots & Indices**

Jan 2020 P2HR - Jan 2020 | P2HR

21 (a) Simplify fully $\dfrac{10x^2 + 23x + 12}{4x^2 - 9}$

....

(3)

$$2^{2y} \times 2^{3y+2} = \frac{8^{5y}}{4^n}$$

(b) Find an expression for n in terms of y.
Show clear algebraic working and simplify your expression.

....

(4)

(Total for Question 21 is 7 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 7

TOPIC: **Algebraic Roots & Indices**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Algebraic Roots and Indices. The tag is acceptable because the index-law part carries the larger share of marks.

Method

For part (a), factorise the numerator and denominator:

$$10x^2 + 23x + 12 = (5x+4)(2x+3)$$

$$4x^2 - 9 = (2x-3)(2x+3)$$

So

$$\frac{10x^2 + 23x + 12}{4x^2 - 9} = \frac{(5x+4)(2x+3)}{(2x-3)(2x+3)}$$
$$= \frac{5x+4}{2x-3}$$

For part (b),

$$2^{2y} \times 2^{3y+2} = 2^{5y+2}$$

Also,

$$\frac{8^{5y}}{4^n} = \frac{(2^3)^{5y}}{(2^2)^n} = 2^{15y-2n}$$

Equate powers of 2:

$$5y + 2 = 15y - 2n$$

$$2n = 10y - 2$$

$$n = 5y - 1$$

FINAL ANSWER

(a) $\frac{5x+4}{2x-3}$, (b) $n = 5y - 1$.

PAST PAPER QUESTION Q#: 21 Marks: 7

TOPIC: **Algebraic Roots & Indices**

Jan 2020 P2HR - Jan 2020 | P2HR

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Show clear algebraic working and simplify your expression.

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(4)

(Total for Question 21 is 7 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 7

Topic: **Algebraic Roots & Indices**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

(a)

$$\frac{10x^2+23x+12}{4x^2-9} = \frac{(5x+4)(2x+3)}{(2x-3)(2x+3)}$$
$$= \frac{5x+4}{2x-3}$$

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$$2^{2y} \times 2^{3y+2} = 2^{5y+2}$$

Also,

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Equate powers:

$$5y+2 = 15y-2n$$

$$2n = 10y-2$$

$$n = 5y-1$$

FINAL ANSWER

(a) $\frac{5x+4}{2x-3}$, (b) $n = 5y-1$

PAST PAPER QUESTION Q#: 22 Marks: 4

TOPIC: **Sequences**

Jan 2020 P2HR - Jan 2020 | P2HR

22 The first term of an arithmetic series S is –6
The sum of the first 30 terms of S is 2865

Find the 9th term of S.

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 4

TOPIC: **Sequences**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

First term:

$$a = -6$$

$$S_{30} = 2865$$

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$2865 = \frac{30}{2}(2(-6) + 29d)$$

$$2865 = 15(-12 + 29d)$$

$$191 = -12 + 29d$$

$$29d = 203$$

$$d = 7$$

The 9th term is

$$u_9 = a + 8d$$

$$u_9 = -6 + 8(7) = 50$$

FINAL ANSWER

50

PAST PAPER QUESTION Q#: 22 Marks: 4

TOPIC: **Sequences**

Jan 2020 P2HR - Jan 2020 | P2HR

22 The first term of an arithmetic series S is –6
The sum of the first 30 terms of S is 2865

Find the 9th term of S.

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 4

TOPIC: **Sequences**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

First term:

$$a = -6$$

$$S_{30} = 2865$$

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

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FINAL ANSWER

50

PAST PAPER QUESTION Q#: 23 Marks: 3

TOPIC: **Completing the Square**

Jan 2020 P2HR - Jan 2020 | P2HR

23 Express $7 - 12x - 2x^2$ in the form $a + b(x + c)^2$ where a, b and c are integers.

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 3

TOPIC: **Completing the Square**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

$$= -2(x^2 + 6x) + 7$$

$$= -2\left((x+3)^2 - 9\right) + 7$$

$$= -2(x+3)^2 + 18 + 7$$

$$= 25 - 2(x+3)^2$$

FINAL ANSWER

$25 - 2(x+3)^2$

PAST PAPER QUESTION

Q#: 23 Marks: 3

TOPIC: **Completing the Square**

Jan 2020 P2HR - Jan 2020 | P2HR

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(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 3

TOPIC: **Completing the Square**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

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FINAL ANSWER

$25 - 2(x+3)^2$

PAST PAPER QUESTION

Q#: 24 **Marks: 5**

Topic: **Coordinate Geometry**

Jan 2020 P2HR - Jan 2020 | P2HR

24 $\mathbf{L}_1$ and $\mathbf{L}_2$ are two straight lines.
The origin of the coordinate axes is O.

$\mathbf{L}_1$ has equation $5x + 10y = 8$
$\mathbf{L}_2$ is perpendicular to $\mathbf{L}_1$ and passes through the point with coordinates (8, 6)

$\mathbf{L}_2$ crosses the x-axis at the point A.
$\mathbf{L}_2$ intersects the straight line with equation $x = -3$ at the point B.

Find the area of triangle AOB.
Show your working clearly.

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

TOPIC: **Coordinate Geometry**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

For L_1,

$$5x + 10y = 8$$

$$10y = -5x + 8$$

$$y = -\frac{1}{2}x + \frac{4}{5}$$

So the gradient of L_1 is $-\frac{1}{2}$.
Since L_2 is perpendicular to L_1, the gradient of L_2 is 2.
Using the point $(8, 6)$:

$$y - 6 = 2(x - 8)$$

$$y = 2x - 10$$

At A, the line crosses the x-axis, so $y = 0$:

$$0 = 2x - 10$$

$$x = 5$$

So $A = (5, 0)$.
At B, $x = -3$:

$$y = 2(-3) - 10 = -16$$

So $B = (-3, -16)$.
Triangle AOB has base $OA = 5$ and perpendicular height 16.

$$\text{Area} = \frac{1}{2} \times 5 \times 16 = 40$$

FINAL ANSWER

40

PAST PAPER QUESTION

Q#: 24 **Marks: 5**

TOPIC: **Coordinate Geometry**

Jan 2020 P2HR - Jan 2020 | P2HR

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Show your working clearly.

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PAST PAPER SOLUTION Q#: 24 Marks: 5

TOPIC: **Coordinate Geometry**

Jan 2020 P2HR - Jan 2020 | P2HR

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FINAL ANSWER

40

PAST PAPER QUESTION Q#: 25 Marks: 3

TOPIC: **Algebraic Proof**

Jan 2020 P2HR - Jan 2020 | P2HR

25 N is a multiple of 5

$A = N + 1$
$B = N - 1$

Prove, using algebra, that $A^2 - B^2$ is always a multiple of 20

(Total for Question 25 is 3 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 3

TOPIC: **Algebraic Proof**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Right-Angled Triangles to Algebraic Proof.

Method

Since N is a multiple of 5, let

$$N = 5k$$

where k is an integer.
Given

$$A = N + 1, \qquad B = N - 1$$

So

$$A^2 - B^2 = (N+1)^2 - (N-1)^2$$

Expand:

$$(N+1)^2 = N^2 + 2N + 1$$

$$(N-1)^2 = N^2 - 2N + 1$$

Therefore

$$A^2 - B^2 = (N^2 + 2N + 1) - (N^2 - 2N + 1)$$

$$= 4N$$

Substitute $N = 5k$:

$$4N = 4(5k) = 20k$$

Since k is an integer, $20k$ is a multiple of 20.

FINAL ANSWER
$A^2 - B^2$ is always a multiple of 20.

PAST PAPER QUESTION

Q#: 25 Marks: 3

TOPIC: **Algebraic Proof**

Jan 2020 P2HR - Jan 2020 | P2HR

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PAST PAPER SOLUTION

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$$= 4N$$

Substitute $N = 5k$:

$$4N = 4(5k) = 20k$$

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FINAL ANSWER
$A^2 - B^2$ is always a multiple of 20.

PAST PAPER QUESTION

Q#: 26 **Marks: 5**

Topic: **Vectors**

Jan 2020 P2HR - Jan 2020 | P2HR

26 The diagram shows trapezium *OACB*.

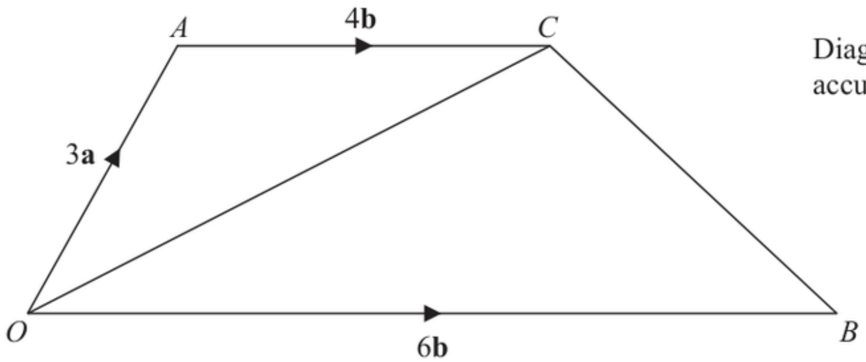


Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 3\mathbf{a}$ $\overrightarrow{OB} = 6\mathbf{b}$ $\overrightarrow{AC} = 4\mathbf{b}$

N is the point on *OC* such that *ANB* is a straight line.

Find $\overrightarrow{ON}$ as a simplified expression in terms of **a** and **b**.

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 5

Topic: **Vectors**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Linear Graphs to Vectors.

Method

$$\vec{OA} = 3\mathbf{a}, \qquad \vec{OB} = 6\mathbf{b}, \qquad \vec{AC} = 4\mathbf{b}$$

So

$$\vec{OC} = 3\mathbf{a} + 4\mathbf{b}$$

Point N lies on OC, so

$$\vec{ON} = t(3\mathbf{a} + 4\mathbf{b})$$

Point N also lies on AB, so

$$\vec{ON} = 3\mathbf{a} + s(6\mathbf{b} - 3\mathbf{a})$$

$$\vec{ON} = (3 - 3s)\mathbf{a} + 6s\mathbf{b}$$

Equate coefficients:

$$3t = 3 - 3s$$

$$4t = 6s$$

So

$$t = 1 - s$$

and

$$t = \frac{3}{2}s$$

$$\frac{3}{2}s = 1 - s$$

$$5s = 2$$

$$s = \frac{2}{5}$$

Then

$$t = \frac{3}{5}$$

Therefore

$$\vec{ON} = \frac{3}{5}(3\mathbf{a} + 4\mathbf{b})$$

$$\vec{ON} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$$

FINAL ANSWER

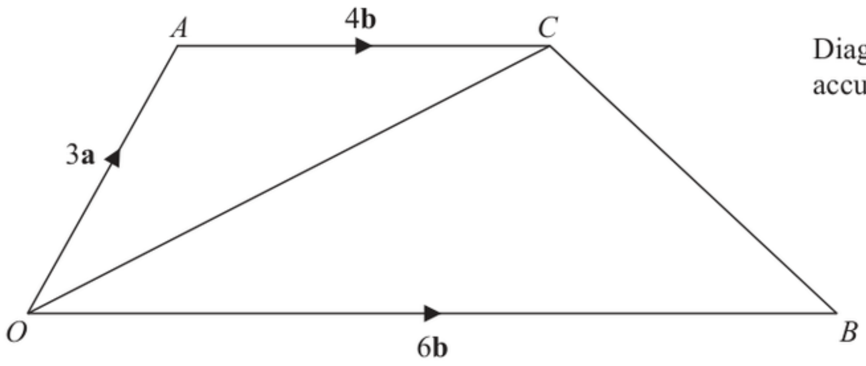
$\vec{ON} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$

PAST PAPER QUESTION Q#: 26 Marks: 5

TOPIC: **Vectors**

Jan 2020 P2HR - Jan 2020 | P2HR

26 The diagram shows trapezium $OACB$.



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PAST PAPER SOLUTION

Q#: 26 Marks: 5

TOPIC: **Vectors**

Jan 2020 P2HR - Jan 2020 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

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Equate coefficients:

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So

$$t = 1 - s$$

and

$$t = \frac{3}{2}s$$

$$\frac{3}{2}s = 1 - s$$

$$5s = 2$$

$$s = \frac{2}{5}$$

Then

$$t = \frac{3}{5}$$

Therefore

$$\vec{ON} = \frac{3}{5}(3\mathbf{a} + 4\mathbf{b})$$

$$\vec{ON} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$$

FINAL ANSWER

$\vec{ON} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$